DAY 9 | POST-READ

Node.js and Express: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
Node.js	A JavaScript runtime with built-in modules (fs, http, path) and the event loop.
Modules	module.exports and require (CommonJS), or export and import (ES modules).
npm	npm init -y, npm install pkg, npm install -D tool, npm run script.
package.json	Name, scripts and dependency ranges. Never commit node_modules.
Lockfile (NEW v4)	package-lock.json records exact versions. Commit it. npm ci installs from it.
npm audit (NEW v4)	Finds known vulnerabilities. --audit-level=high fails CI. Test after npm audit fix.
Express	express(), app.use(express.json()), routers, controllers, res.status().json().
Request data	req.params (path), req.query (query string), req.body (JSON), req.get (headers).
Async errors	Express 5 forwards rejected promises to the error middleware (err, req, res, next).
Postman	Requests, JSON bodies, {{baseUrl}} environments, pm.test checks, collection runs.

Part B: Going beyond today

1. nodemon for auto reload

npm install -D nodemon, then add "dev": "nodemon src/server.js" to scripts. It restarts the server on every save, like uvicorn --reload.

2. Environment variables

```
const PORT = process.env.PORT || 3000;
// Secrets come from the environment or a git-ignored .env file, never from the code.
```

3. Validation libraries

Instead of many if checks, a schema library validates the whole body at once, similar to Pydantic. Check the package carefully before adding it, as you learned today.

4. Overrides for a vulnerable sub-dependency

Sometimes the vulnerable package is deep in the tree and the top package has not released a fix yet. npm "overrides" in package.json can force a fixed version. Test well, and remove the override once the top package updates.

5. Testing Express without Postman

Start the app on a random port in a test script and call it with fetch, or use a test library. On Day 15 you will automate API tests for FastAPI with PyTest; the same idea applies here.

Part C: Check yourself

- Why should node_modules not be committed to Git, but package-lock.json should?
- What is the difference between npm install and npm ci?
- What does ^5.1.0 allow that 5.1.0 does not?
- Where do req.params, req.query and req.body come from?
- What happens in Express 5 when an async route throws an error?

Part D: Practice ideas for tonight

- Add PATCH and filtering by done=true or false to your Express Tasks API.
- Add Postman tests to every request in your collection and run the whole collection.
- Run npm audit on your project and write one line per finding in the vulnerability report template.

Coming up on Day 10: React basics: single page apps, JSX, components, props, state and events, building the SmartDesk dashboard with Codex, and Basics assessment 2.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today